

Gas absorption:

Fundamentally, there are two mass transfer processes in gas/liquid absorption: mass transfer from the bulk air to the air/water interface, and mass transfer from the air/water interface to the bulk water. See p 547 in Unit Ops textbook for more information.

We can't measure individual mass transfer coefficients from those two processes since we don't know the carbon dioxide concentrations at the air/water interface or the interfacial area. We can get around this by describing mass transfer from the bulk air to the bulk water using an absorption rate equation based on the mass transfer flux equation:

$$N_a = \text{Mass transfer rate} = K_a ([\text{CO}_2]_{\text{air, out}} - [\text{CO}_2]_{\text{water, out}})$$

$$\text{Mass transfer rate} = F_{\text{water}} ([\text{CO}_2]_{\text{water, out}} - [\text{CO}_2]_{\text{water, in}})$$

$$\text{Mass transfer rate} = F_{\text{air}} ([\text{CO}_2]_{\text{air, out}} - [\text{CO}_2]_{\text{air, in}})$$

Where

F_{water} = water flow rate (volume /time)

(K_a) = overall volumetric mass transfer coefficient (contains the area since we can't calculate it)

$[\text{CO}_2]_{\text{water, out}}$ = carbon dioxide concentration from titrations

$[\text{CO}_2]_{\text{water, in}}$ = carbon dioxide concentration in water initially, from titrations

$[\text{CO}_2]_{\text{air, out}}$ = carbon dioxide concentration from Hempl measurement

$[\text{CO}_2]_{\text{air, in}}$ = initial carbon dioxide concentration from CO_2 flowrate

We can use the second equation or third equation to calculate the mass transfer rate, and then first equation to calculate the mass transfer coefficient.

In a way, this is similar to the mass transfer calculations that you did last semester for the membrane air separation where we did not know the interfacial area of the hollow fibers or the O_2 and N_2 concentrations at the air/fiber inlet and outlet interfaces, so we lumped together the mass transfer coefficient and the interfacial area.